The Geopolitics of Food and Fertilizer

Weaponization, Crisis, and the Rise of Agritech Diplomacy

A Case Study Report in Geopolitics and Agricultural Economics

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Independent analytical report. Figures are rounded; see Appendix for data notes.

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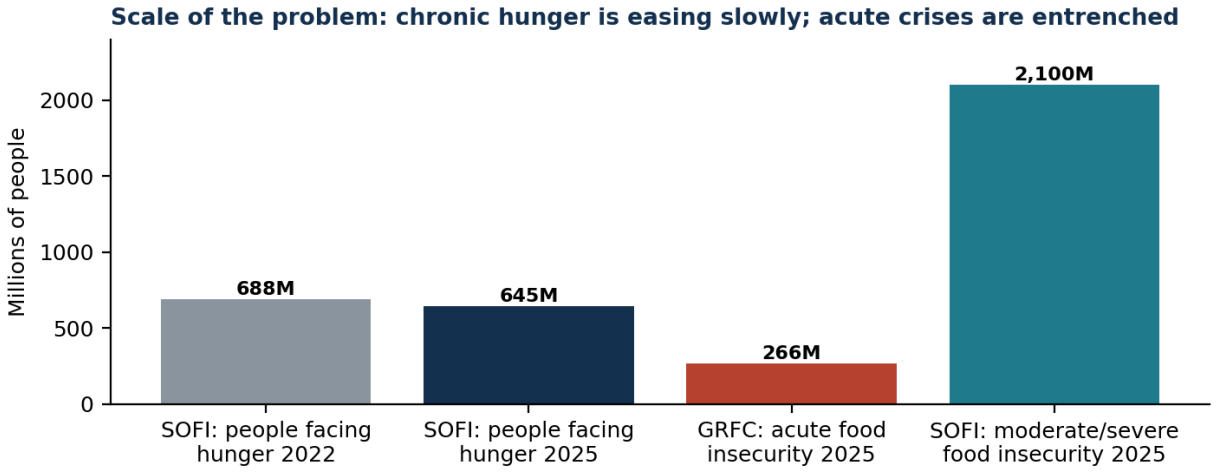
1. Executive Summary

For most of the post-1945 era, grain, oilseeds and fertilizer were treated as ordinary commodities: bulky, cheap, and governed by weather and price. That assumption no longer holds. Between 2020 and 2026, a pandemic, the full-scale invasion of Ukraine, a wave of export bans, and the 2026 closure of the Strait of Hormuz have shown that the inputs and outputs of farming are strategic levers that states use to project power, punish rivals and create dependencies. Wheat is concentrated in a few exporters, nitrogen is tied to natural gas and a few Gulf and Eurasian producers, phosphate to China and Morocco, and potash to Canada, Russia and Belarus. Each concentration is a potential choke point.

The record is consistent. The 1980 U.S. grain embargo, the 1960s use of PL-480 food aid against Egypt, China's 2020 barley duties on Australia, India's 2022-23 export bans and Russia's conditional approach to the Black Sea Grain Initiative all show the same logic: those who control supply set the terms. The Middle East, the world's most import-dependent region, absorbs the shock first. Egypt, Lebanon and Yemen illustrate how a distant price spike becomes currency stress, subsidy strain and political risk.

The humanitarian stakes are high. The UN's SOFI 2026 estimates 645 million people faced hunger in 2025, a third annual decline but far from the 2030 goal, while the Global Report on Food Crises 2026 counts 266 million in acute food insecurity and two confirmed famines in a single year. Legal protections for the right to food exist, but enforcement is weak. Climate change will widen the gap by intensifying drought, heat and water scarcity in the same regions that import most of their food.

This report argues that lasting protection comes from reducing the *substitutability gap*, not from hoping exporters behave. **Agritech Diplomacy** does this: wealthy and technologically advanced states export irrigation, controlled-environment agriculture, soil and digital tools, and local input production to vulnerable partners. Hydroponics and aquaponics will not replace wheat or rice, and their energy and capital demands are real, but they can localise fresh produce and protein, cut import bills, and demonstrate a model that, combined with precision fertilizer use and regional input production, weakens coercive leverage.



Sources: SOFI 2026 (Jul 2026): 645M hungry in 2025, 43M fewer than 2022 peak; GRFC 2026 (Apr 2026). 2022 value derived (645+43).

Figure 1. Hunger and food-crisis headline figures (SOFI 2026; GRFC 2026).

2. Problem Statement: The Weaponization of Agriculture

2.1 Why food and fertilizer are unusually potent levers

Food is the most inelastic of all goods: households cannot postpone eating, and governments know that bread prices can topple regimes. Three structural features turn this vulnerability into leverage. First, **concentration**: a few exporters supply the bulk of traded wheat, maize, rice, vegetable oils and key fertilizers. Second, **low short-run substitutability**: importers cannot quickly grow or source an alternative because of agro-climatic limits, storage and logistics, and the fixed cycle of planting seasons. Third, **thin buffers**: most import-dependent states hold weeks, not months, of strategic cover and have limited foreign-exchange reserves.

Fertilizer sits one layer upstream and is therefore a second-order lever. Synthetic nitrogen made by the Haber-Bosch process is estimated to support roughly half of the world's population (Smil), and phosphate and potash have no substitutes. A restriction on fertilizer does not hurt this season's meals; it hurts the next harvest, at a lag of six to eighteen months, which complicates both diagnosis and response.

2.2 Instruments of agricultural coercion

Instrument	How it works	Illustrative case
Export ban / quota / duty	Exporter withholds supply or taxes it to protect domestic prices or to punish a buyer	India wheat (2022), non-basmati rice (2023); Indonesia palm oil (2022); Russia floating wheat duty
Embargo or sanction	Target-specific denial of grain, aid or inputs	U.S. grain embargo on USSR (1980)
Import ban / tariff as retaliation	Importer closes market to a rival's farmers	China-Australia barley (2020-23); China-Philippines bananas (2012)
Aid conditionality	Food aid is delivered, delayed or cancelled depending on politics	PL-480 and Egypt in the 1960s
Blockade / port attack	Physical denial of shipping or export infrastructure	Black Sea ports 2022-23; Odesa and Izmail strikes
Corridor conditionality	Safe passage tied to unrelated demands	Black Sea Grain Initiative (2022-23)
Input monopoly pricing	State-linked producers throttle or re-route fertilizer or energy	Gas-linked nitrogen shock 2021-22; Belarus potash
Regulatory pretexts	Sanitary, inspection or licensing barriers used selectively	China phosphate and urea export inspections (2021 onwards)

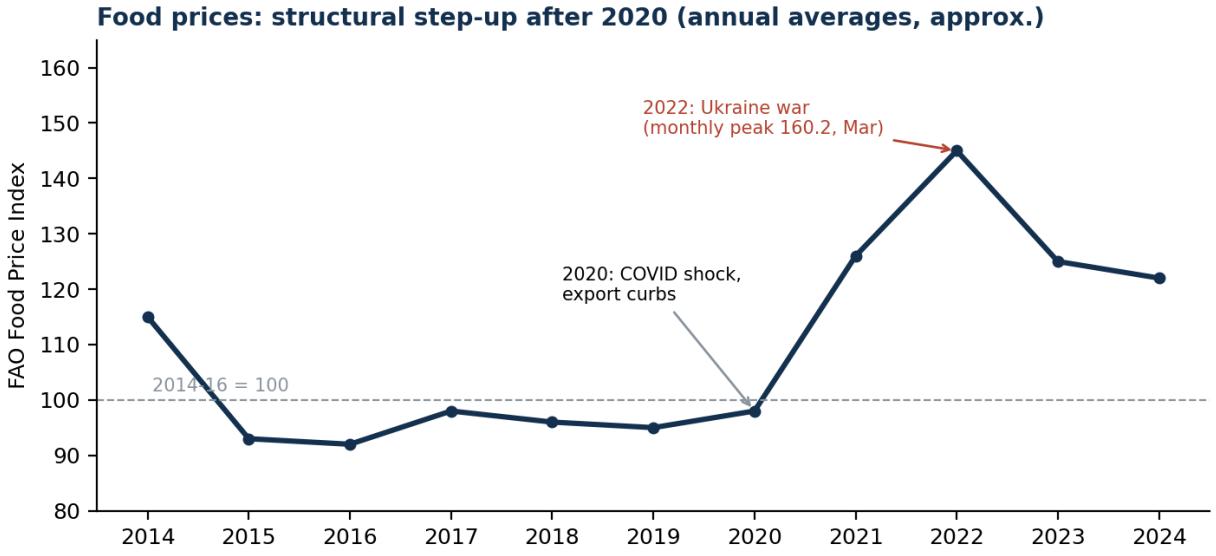
2.3 Food choke points

A **food choke point** is a geographic or institutional node through which a large share of food or input trade must pass, where disruption has outsized, hard-to-reroute effects. Maritime examples include the Turkish Straits (Bosphorus and Dardanelles), which carry Black Sea grain; the Strait of Hormuz, which handles a large share of Gulf nitrogen, sulfur and LNG; the Suez Canal and Bab el-Mandeb, which link Black Sea and European cargo to Asia and East Africa; the Panama Canal, whose 2023-24 drought restricted U.S. grain flows to Asia; and the Strait of Malacca. Non-maritime choke points include a handful of fertilizer pipelines and rail corridors, single-country phosphate supply, port silos such as Beirut's before 2020, and payment and insurance systems. Chinese and Russian state traders, and Gulf and Indian state purchasing agencies, add an institutional choke point: a political decision can redirect cargoes overnight.

2.4 Supply chain manipulation

Manipulation rarely requires a formal embargo. Typical techniques include: (i) **timing** restrictions at planting or tender dates; (ii) **discretionary licensing** that rewards friendly buyers; (iii) **price signalling** through state traders; (iv) **gift diplomacy**, such as donated grain or fertilizer that builds

goodwill and dependence; (v) **financial friction**, including reduced access to payments, insurance and shipping finance; and (vi) **cascading export restrictions**, in which one country's ban triggers panic bans elsewhere. IFPRI's tracker found export restrictions covered roughly 17% of globally traded calories at the 2022 peak, a level comparable to the 2007-08 crisis.



Source: FAO Food Price Index; values rounded by author. Verify exact series at fao.org.

Figure 2. FAO Food Price Index, annual averages (approximate, rounded). Prices have settled well above their pre-2020 range.

3. Historical and Modern Cases of Food Diplomacy

The use of food as a political instrument is ancient, from sieges to colonial famine policy, but the modern record reveals four recurring patterns: great-power denial, aid conditionality, retaliatory trade measures, and food as a weapon of war.

3.1 Great-power denial: the 1973 soybean and 1980 grain embargoes

In June 1973, facing domestic price inflation, the United States imposed temporary export controls on soybeans. Japan, which relied heavily on U.S. beans, experienced the move as a shock and responded by diversifying toward Brazil, a decision that helped create the Brazilian soybean powerhouse. In January 1980, following the Soviet invasion of Afghanistan, President Carter suspended roughly 17 million tonnes of grain sales to the USSR. The embargo's effect was limited because Argentina, Canada, the EU and others filled most of the gap, and it imposed heavy costs on U.S. farmers. President Reagan lifted it in April 1981. The lessons are enduring: unilateral denial works only when the exporter is dominant and the buyer cannot substitute, and it can permanently erode the exporter's market share.

3.2 Aid as leverage: PL-480, Egypt and Bangladesh

U.S. Public Law 480 ('Food for Peace', 1954) turned surplus disposal into diplomacy. In the mid-1960s, the Johnson administration shortened Egypt's PL-480 wheat shipments to a 'short leash' as President Nasser's foreign policy diverged from Washington's. Egypt, which had come to rely on U.S. wheat, was pushed toward Soviet supply and later toward diversified import sourcing. In 1974, shipments to Bangladesh were delayed during a famine period after Dhaka sold jute to Cuba; historians debate how much the delay contributed to the famine, but the episode became a cautionary tale about politically conditioned aid. Aid-based dependence shapes the policies of recipients for decades; many later invested in domestic agricultural policy specifically to avoid it.

3.3 Retaliatory trade measures

Case	Year	Mechanism	Outcome
China-Philippines bananas	2012	Phytosanitary inspections slowed Philippine banana entry during the Scarborough Shoal standoff	Fruit rotted at port; Manila diversified exports but the signal was clear
Russia food embargo	2014	Counter-sanctions on EU, U.S. and others' food after Crimea	Domestic import substitution; Russia emerged as a larger grain exporter
China-Australia barley	2020-23	Anti-dumping duties of about 80% on barley, plus wine and other measures	Australian exports redirected; China lifted barley duties in 2023 after diplomatic thaw
India wheat ban	2022	Export ban after heatwave and price spike	Importers lost an expected supplier; global prices jumped
India rice curbs	2023-24	Non-basmati white rice ban, later relaxed	African and Asian importers faced price shocks; later normalized
Indonesia palm oil	2022	Temporary export ban to lower domestic cooking-oil prices	Global edible-oil prices spiked, then fell after ban was lifted

3.4 Food as a weapon of war

From the World War I 'hunger blockade' of Germany to the siege of Leningrad, the Biafran blockade (1967-70) and the Ethiopian famine of 1984-85, deliberate denial of food to civilians has recurred throughout modern conflict. The Soviet Holodomor of 1932-33 is the most consequential case of food used against a domestic population. More recently, the GRFC 2026 reports that famine (IPC Phase 5) was confirmed in parts of Gaza and Sudan in 2025, the first time two famines were confirmed in one year in the report's history, with conflict identified as the primary driver. Restrictions on humanitarian access, attacks on markets and farms, and the siege of cities remain the operative tools.

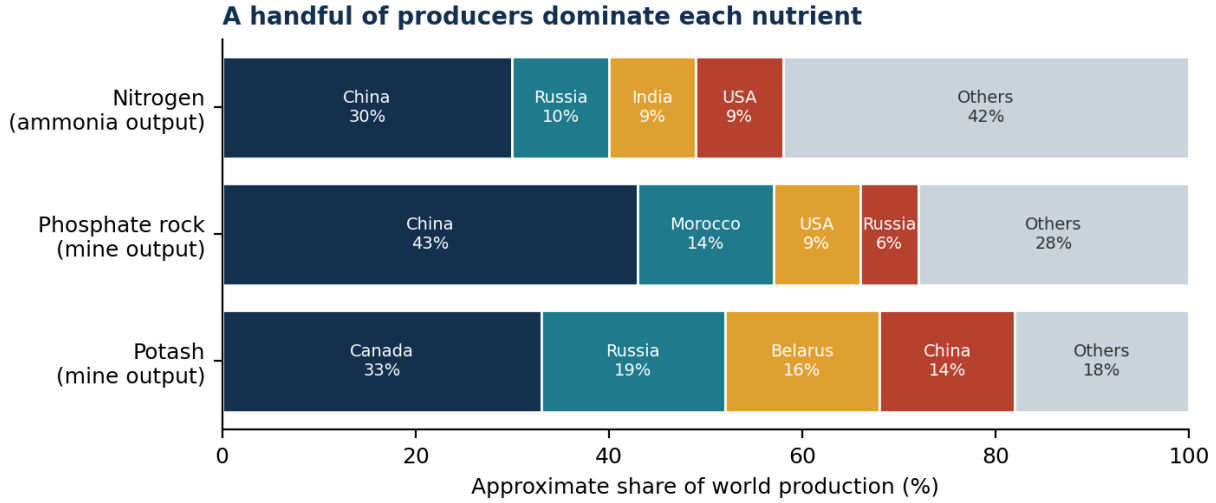
3.5 Patterns and lessons

- **Leverage depends on market share and substitutes.** The 1980 embargo failed; blockades of landlocked or insular buyers succeed.
- **Short-term pain, long-term loss.** Importers diversify (Japan to Brazil; Egypt to Black Sea), and the coercer's share declines.
- **Domestic politics drive many 'weaponizations'.** Many export bans are inward-looking price-control measures whose external harm is a side effect, yet the effect on importers is identical.
- **Humanitarian carve-outs are fragile.** Food is nominally exempt from most sanctions regimes, but financing, shipping and insurance frictions make exemptions partial in practice.

4. Historical and Modern Cases of Fertilizer Diplomacy

4.1 Concentration of the three macronutrients

Nitrogen (N) is produced from atmospheric nitrogen and natural gas (or coal in China); gas is typically 70-90% of ammonia's cash cost, so access to cheap gas is the competitive advantage. Leading producers include China, Russia, India, the United States and Gulf states such as Qatar, Saudi Arabia and Iran. **Phosphate (P)** is mined; China is the leading producer, while Morocco, through the state-owned OCP, holds roughly 70% of the world's phosphate-rock reserves according to USGS. **Potash (K)** is mined from a few deep deposits: Canada, Russia and Belarus together account for more than two-thirds of output.



Source: USGS Mineral Commodity Summaries (recent years), rounded; indicative. Morocco also holds ~70% of phosphate reserves.

Figure 3. Approximate production shares by nutrient (USGS-based, rounded).

4.2 Case studies

Belarus potash and the 2021 sanctions

After the 2020 election crackdown, the EU (June 2021) and the United States (August 2021) imposed sanctions that hit Belaruskali and related potash exports. Lithuania ended transit through Klaipeda in 2022, forcing rerouting through Russian ports. The episode showed that potash buyers in Brazil, India and Southeast Asia, who had no stake in the political dispute, absorbed the price and logistics costs; it also deepened Belarus's reliance on Russian corridors.

China's export restrictions, 2021 onward

In late 2021, with domestic prices rising, China tightened inspection and export rules for urea and phosphates. Export volumes fell sharply and global prices rose. China has since adjusted quotas repeatedly. In 2026, Beijing relaxed urea exports to an allowance of roughly 5-5.5 million tonnes, which helped cool the market after the Hormuz shock. Dependents such as India and Pakistan treat Chinese policy as a variable they cannot control, which has been reflected in reported curbs on specialty fertilizers to India.

Russia, gas and the 2021-22 nitrogen shock

European gas prices surged in 2021 and again in 2022 as Russian pipeline flows were cut, forcing ammonia plants in Europe to curtail output and pushing urea above US$900 per tonne in spring 2022. Russia is the world's largest fertilizer exporter by volume, and although fertilizers were formally excluded from most Western sanctions, payment, shipping and insurance frictions, and the closure of the Togliatti-Odesa ammonia pipeline, reduced effective availability. Russia then used donated fertilizer, including cargoes stranded in EU ports and offers to African states, as a diplomatic gift. The EU subsequently adopted phased tariffs on Russian and Belarusian fertilizers

(2025) to reduce its own exposure.

Morocco, OCP and the politics of phosphate

OCP's dominance gives Morocco leverage and visibility across Africa through fertilizer supply, investment and training. The origin of some phosphate from Western Sahara has led to repeated legal disputes in EU courts and to shipment seizures, which demonstrates that fertilizer supply chains are entangled with territorial disputes.

Sri Lanka 2021: self-inflicted dependency

Sri Lanka's abrupt 2021 ban on synthetic fertilizer imports, with the aim of going organic nationwide, was reversed within months after rice output fell significantly (estimates around 20%) and tea harvests suffered. It remains the standard warning that abandoning inputs without a tested substitute is itself a self-inflicted shock, and it informs the sober recommendations in Section 10.

Box A. The 2026 Hormuz fertilizer shock: a live stress test

What happened. After the war involving the United States, Israel and Iran began on 28 February 2026, commercial traffic through the Strait of Hormuz collapsed. Analysts described about one-third of global fertilizer trade as exposed; the WTO-AXSMarine tracker puts Hormuz's pre-war share of traded nitrogen near 20%, plus a large share of sulfur, an input for phosphate. Figures differ by benchmark and definition.

Prices. Urea roughly doubled in weeks by some measures (IFPRI), with Middle East export prices around US$700 per tonne in March (Argus via Reuters). The World Bank (June 2026) projected a 31% rise in fertilizer prices for 2026, with urea up 46% month-on-month at one point.

Where it stands. Despite an 8 April ceasefire and a June Iran-Oman shipping MoU, WTO data through early September 2026 show fertilizer shipments through the strait still near zero. Urea had eased to roughly US$443 per tonne by 4 September because China widened export quotas and Hormuz talks progressed. IFPRI authors warn that a prolonged closure could extend the crisis into 2028.

Who is exposed. India sources over 40% of its urea and phosphate fertilizers from the Middle East; Brazil imports nearly all its urea, about half of it via Hormuz. Mercy Corps documents knock-on effects in Sudan, Somalia, Ethiopia, Pakistan, Myanmar and Lebanon.

Analytical point. Unlike 2022, this shock was not a deliberate export weapon but a conflict-driven choke-point closure. The effect on dependent farmers is the same, which is why resilience, not intent, should be the policy target.

5. Deep-Dive: Russia-Ukraine Wheat Diplomacy and the Middle East

5.1 The conflict and the Black Sea supply chain

Before February 2022, Russia and Ukraine together supplied roughly 30% of world wheat exports, a substantial share of maize and barley, and around half or more of global sunflower oil trade. Ukraine's ports on the Black Sea (Odesa, Chornomorsk, Pivdennyi) moved the bulk of its exports. The invasion and naval blockade stranded an estimated 20 million tonnes or more of grain, mined approaches, and removed Ukraine from the market at the height of post-pandemic tightness. Chicago wheat futures reached their highest levels in over a decade in March 2022, and the FAO Food Price Index hit its record monthly high of 160.2.

5.2 The strategic lever: the Black Sea Grain Initiative

Signed in Istanbul on 22 July 2022 and brokered by Turkey and the UN, the Black Sea Grain Initiative (BSGI) created a Joint Coordination Centre to inspect cargoes and allow Ukrainian ships to pass through a protected corridor. Roughly 33 million tonnes were exported before Russia withdrew on 17 July 2023. A parallel memorandum committed the UN to help remove obstacles to Russian food and fertilizer exports. This structure created a leverage mechanism: Russia could slow inspections, delay renewals (the deal was extended several times, with the later extensions shortened to 60 days) and link continuation to demands including reconnecting Rosselkhozbank to SWIFT, spare parts, insurance access and the ammonia pipeline.

Russia's broader wheat diplomacy complemented the corridor. It consolidated exports through large state-aligned traders, used export quotas and a floating duty to manage domestic prices, offered discounted long-term supply to partners such as Egypt and Turkey, and announced free grain deliveries of up to 25,000-50,000 tonnes each to six African countries at the July 2023 Russia-Africa summit. At the same time, strikes on Odesa and Danube ports after the BSGI collapse raised Ukraine's costs. Ukraine responded by opening its own Black Sea corridor in August 2023 and increasing Danube and overland routes. By 2024 Ukrainian exports had substantially recovered, but at higher logistics and insurance costs.

5.3 Impact on the Middle East and North Africa

The MENA region is the world's largest net importer of cereals and the most water-scarce region, and its states have long used subsidized bread as a pillar of the social contract. The price spike of 2022 therefore transmitted through three channels: **import bills** (higher price and freight), **currency and fiscal stress** (dollar-denominated grain, subsidized retail prices), and **social stability** (the political memory of the 1977 Egyptian bread riots and the 2008-11 price spikes that preceded the Arab Spring).

Country	Vulnerability profile	2022-26 transmission
Egypt	World's largest wheat importer (roughly 12-13 Mt a year); before 2022 about 80% of imports came from Russia and Ukraine; subsidized 'aish baladi' bread reaches tens of millions	Import-bill surge; pound devalued sharply in 2022-24; IMF support and large Gulf investment (e.g., 2024 UAE Ras El-Hekma deal); state agency took over wheat purchases; diversified to Russia (direct deals), EU, Romania, Argentina and others
Lebanon	Imports nearly all wheat; Ukraine was the main source (about half or more); Beirut port silos, with most grain-storage capacity, destroyed in the August 2020 explosion	Price shock on top of financial collapse (since 2019); thin stocks; later conflict and displacement and 2026 fuel-price pressure (Mercy Corps) compound stress
Yemen	Imports about 90% of staple foods; war-damaged ports; reliant on WFP; Russia and Ukraine major wheat suppliers	Price spike hit a country already near famine thresholds; Red Sea shipping risks since late 2023 raised costs; aid funding declines in 2025-26 reduce coverage
Tunisia, Libya, Syria	High wheat-import dependence on Black Sea; fiscal fragility	Subsidy strain, queue and shortage episodes, reliance on emergency finance

Country	Vulnerability profile	2022-26 transmission
Gulf (GCC)	Import 80%+ of food but have financial buffers and storage	Absorbed shock; accelerated outbound farmland deals and domestic agritech investment; now exposed to Hormuz risk itself

Assessment. The Black Sea episode did not produce a classic embargo. It produced *conditional access*, where each importer's exposure was determined by logistics, credit and diplomacy. Russia benefitted commercially from high prices and politically from visible 'rescue' gestures, while importers learned that diplomatic positions on Ukraine could influence supply terms. Importers in the Middle East largely maintained pragmatic relationships with both Moscow and Kyiv; the real cost fell on fiscal balances and on the poorest households.

6. The Global Right to Food and Institutional Defenses

6.1 Legal frameworks

Instrument	Provision	Relevance to weaponization
Universal Declaration of Human Rights (1948), Art. 25	Right to a standard of living adequate for health and well-being, including food	Moral and customary foundation; not directly enforceable
ICESCR (1966), Art. 11	Right to adequate food and the fundamental right to be free from hunger; states must cooperate internationally, including ensuring equitable distribution of world food supplies relative to need	Creates obligations of conduct, including extraterritorial duties of respect and cooperation (CESCR General Comment 12, 1999)
FAO Voluntary Guidelines on the Right to Food (2004)	Practical guidance for national implementation	Soft law; shaped national constitutions and strategies
Additional Protocol I (1977), Art. 54	Prohibits starvation of civilians as a method of warfare and attacks on objects indispensable to survival	Binding in international armed conflicts; reflects customary law
Rome Statute, Art. 8(2)(b)(xxv); 2019 amendment	War crime of intentionally using starvation of civilians as a method of warfare; extended to non-international conflicts in 2019	Provides a route to prosecution; few cases to date
UN Security Council Resolution 2417 (2018)	Condemns starvation as a method of warfare; links conflict and hunger; asks Secretary-General to report risks of famine	Early-warning and accountability signal; implementation inconsistent
WTO Agreement on Agriculture, Art. 12; Marrakesh Decision (1994)	Requires notice and 'due consideration' of importers' food security when imposing export prohibitions	Weak; no effective sanction; MC12 (2022) exempted WFP humanitarian purchases from export bans

The central weakness is clear. **Rights are individual and states' duties are diffuse**: no binding treaty prohibits an exporter from restricting commercial food or fertilizer sales to another state, and the WTO's disciplines on export restrictions remain largely hortatory.

6.2 What the institutions are saying

Organization	Recent finding or warning
FAO, IFAD, UNICEF, WFP, WHO (SOFI 2026, July 2026)	645 million people (7.8%) faced hunger in 2025, a third consecutive annual decline and 43 million fewer than the 2022 peak; about 2.1 billion faced moderate or severe food insecurity; Africa (about 309 million) overtook Asia (about 292 million) as the region with the most hungry people; around 510-520 million could still be hungry in 2030; the report warns that the 2026 Middle East conflict, weather extremes and aid cuts threaten progress
Global Network Against Food Crises (GRFC 2026, April 2026)	About 266 million people in 47 countries and territories in acute food insecurity in 2025; famine confirmed in parts of Gaza and Sudan; conflict the primary driver (over 147 million in conflict-driven contexts); data gaps and falling humanitarian funding are themselves risks
World Food Programme	Funding declines force prioritization: WFP plans to reach about 110 million people in 2026, roughly a third of those in need
World Bank (Commodity Markets Outlook, June 2026)	Fertilizer prices projected up about 31% on average in 2026; the Hormuz disruption lifted agricultural price indices by about 8%; warns of an affordability-driven crisis
IFPRI (Glauber, Laborde, Arita and others)	Export restrictions covered about 17% of traded calories in 2022; documented how clusters of bans amplify price spikes; 2026 analysis warns that prolonged Hormuz closure could extend the fertilizer crisis into 2028
UN Secretary-General (GRFC 2026 foreword)	Called the confirmation of two famines in one year an unprecedented development

6.3 Attempts to insulate food from geopolitical conflict

- **Black Sea Grain Initiative (2022-23)** showed that a humanitarian corridor can work under a neutral broker (Turkey and the UN), but also that it is only as durable as the weakest party's consent.

- **WTO MC12 decision (June 2022)** exempted WFP humanitarian purchases from export restrictions: narrow but a rare binding step.
- **AMIS (2011)**, the G20-backed market information system, improves transparency and reduces panic buying.
- **Regional reserves** such as ASEAN's APTERR rice reserve and ECOWAS's regional food security reserve diffuse national hoarding incentives.
- **Financing tools**: the IMF's 2022 Food Shock Window and World Bank crisis facilities that help importers pay bills without rationing.
- **Africa Fertilizer and Soil Health Summit (Nairobi, 2024)** set targets for regional input supply and soil health.
- **Global Alliance Against Hunger and Poverty (launched under Brazil's 2024 G20 presidency)** mobilizes finance and policy sharing.
- **Reform proposals** include: binding disciplines on food and fertilizer export bans; a standing humanitarian carve-out for food, fertilizer, insurance and payment channels in sanctions and conflict; an international emergency grain and fertilizer reserve; and formal recognition of attacks on food systems as accountable conduct.

Assessment. The architecture is stronger on monitoring and relief than on prevention. Resilience at the national and regional level therefore has to carry more of the burden than multilateral rules currently can.

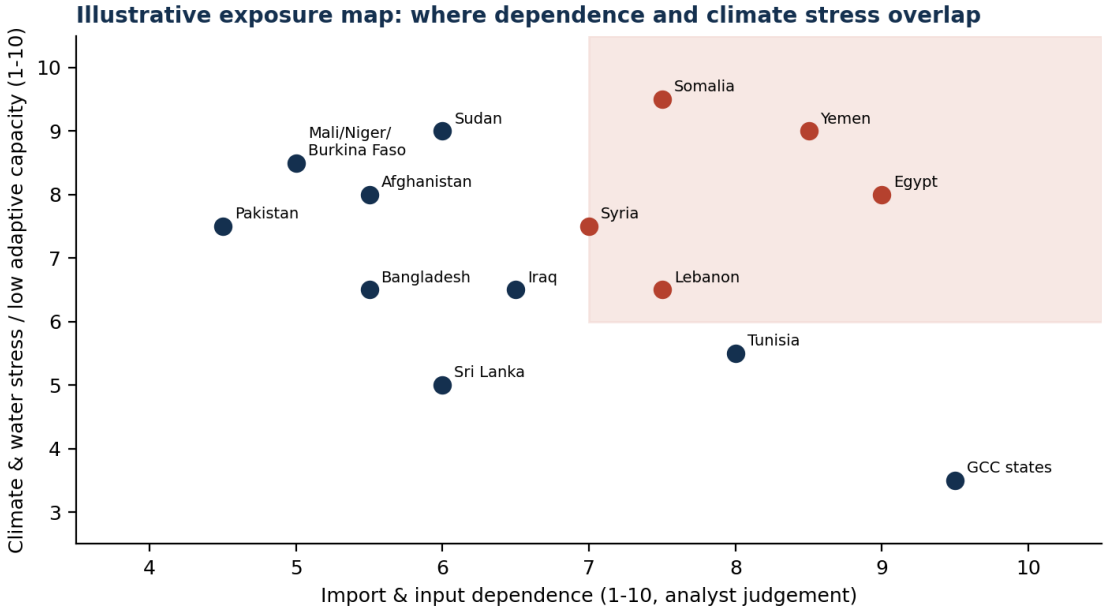
7. Future Outlook, Climate Change and Vulnerable Nations

7.1 How climate will amplify the vulnerabilities

The IPCC's Sixth Assessment Report finds with high confidence that climate change has already slowed agricultural productivity growth and that risks to yields rise with warming, with the largest negative effects concentrated in low latitudes. Over the next two decades the main pressures are:

- **Heat and drought.** Heatwaves in India in 2022 reduced wheat yields and helped trigger the export ban; the Horn of Africa's 2020-23 drought, the worst in about four decades, left millions facing acute hunger.
- **Water scarcity.** Agriculture accounts for roughly 70% of global freshwater withdrawals; WRI's Aqueduct analysis places a majority of the world's most water-stressed countries in the Middle East and North Africa. Competition over transboundary rivers (Nile, Tigris-Euphrates, Indus) will intensify.
- **Desertification and soil degradation.** The Sahel, Central Asia and the Mediterranean rim face rising aridity and salinization.
- **Sea-level rise and salinization.** The Nile Delta, Bangladesh and low-lying small island states risk loss of farmland.
- **Volatility and simultaneous failures.** Correlated shocks in multiple breadbaskets could trigger multi-country export restrictions: the exact condition under which coercion peaks.
- **Energy and input shocks** such as the 2026 Hormuz crisis add price risk independent of weather.

7.2 Most exposed countries and regions



Qualitative author scoring informed by FAO, WFP, WRI Aqueduct, ND-GAIN; not a published index. Shaded zone = highest coercion risk.

Figure 4. Illustrative exposure map. Scores are the author's qualitative judgements, informed by public indicators, and are meant to prompt discussion rather than rank countries precisely.

Region / country	Principal vulnerabilities	Coercion pathway
Egypt, Lebanon, Tunisia, Libya, Syria, Iraq	Heavy cereal imports, water scarcity, fiscal strain	Wheat access and price; Nile flows; Gulf and IMF financing as soft conditionality
Yemen, Somalia, Sudan, South Sudan	Conflict, drought, aid dependence, port fragility	Aid access, port control, funding cuts, input prices

Region / country	Principal vulnerabilities	Coercion pathway
Sahel (Mali, Niger, Burkina Faso, Chad)	Desertification, conflict, thin markets, reliance on imported rice and wheat	Gift grain diplomacy; fertilizer access; shifting security partners
South Asia (Pakistan, Bangladesh, Afghanistan, Sri Lanka)	Heat, floods, glacier-fed water, imported fertilizer, energy import dependence	Gulf and Chinese input supply; Indian rice and wheat policy
Gulf states	Import 80%+ of food; extreme water scarcity; Hormuz exposure	Chokepoint risk; reliance on a few suppliers
Small island states and Central America / Caribbean (Haiti)	Climate shocks, import reliance, limited storage	Freight and finance; U.S. and Latin American export policy

Outlook. SOFI 2026 projects about 510-520 million people could still be hungry in 2030, with a growing share in Africa. If climate shocks coincide with another chokepoint closure or export-ban cascade, those numbers could be exceeded. The practical conclusion is that exposure to a single supplier, route or input is the variable policymakers can change.

8. The Solution: Agritech Diplomacy

8.1 Definition

Agritech Diplomacy is a foreign policy and development approach in which states, development banks and firms deliberately export agricultural technology, knowledge, finance and standards to partner countries in order to raise those partners' domestic production, input self-sufficiency and climate resilience, while building long-term commercial and political relationships. It extends food-aid diplomacy by shifting the offer from *commodities* to *capabilities*, and extends classical development by treating food security as a component of strategic autonomy.

8.2 The logic: leverage equals dependence times non-substitutability

An exporter's leverage over an importer rises with the importer's dependence on that exporter and falls with the importer's ability to substitute. Agritech Diplomacy attacks the second term. Every tonne produced domestically, every alternative supplier qualified, every month of buffer stock and every locally produced input raises the price a coercer would have to pay for using the lever, and often renders the threat not credible.

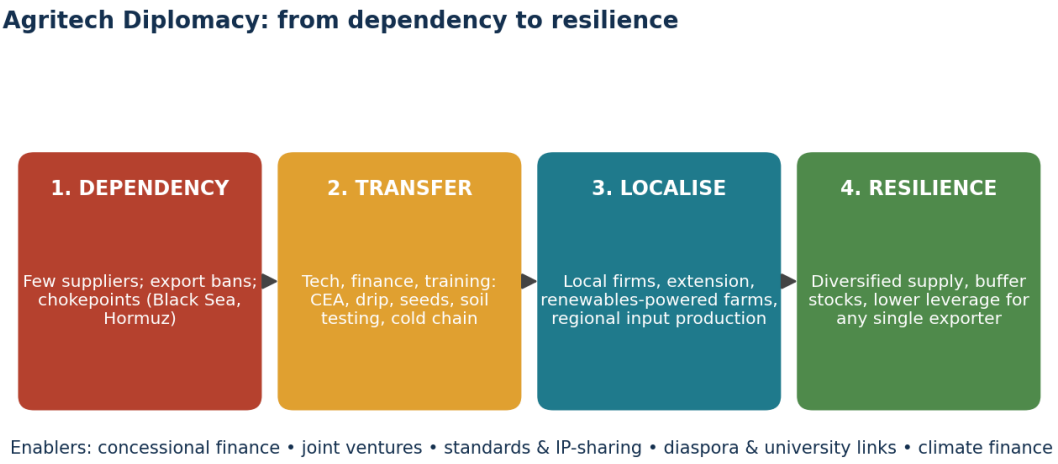


Figure 5. The agritech diplomacy pathway.

8.3 Pillars

Pillar	What it involves	Examples of practice
Technology transfer	Drip and precision irrigation, climate-smart seed, soil testing, greenhouse and controlled-environment systems, cold chains, digital advisory	Israel's MASHAV; Dutch agri-food partnerships; Japan's JICA CARD rice initiative in Africa; China's agricultural technology demonstration centers; Korea's K-Food Belt programs
Finance and risk-sharing	Concessional loans, blended finance, guarantees, insurance, green-ammonia and renewable-energy co-investment	World Bank, IFAD and regional development banks; Gulf sovereign funds; reported Indian long-term green-ammonia offtake deals
Local input production	Regional fertilizer blending and production, biofertilizers, organic-mineral blends, soil-health programs	Morocco's OCP in Africa; Nigeria's large urea plants; Nairobi 2024 summit commitments
Capacity and standards	Extension systems, training, research partnerships, harmonized phytosanitary and quality standards	University links, Wageningen-type centers, South-South technical cooperation
Data and early warning	Weather, soil and market data; satellite monitoring; open platforms	AMIS, FEWS NET-type tools, national digital agriculture platforms

8.4 Design principles and risks

- **Avoid replacing one dependency with another.** If a donor's proprietary seeds, platforms or spare parts lock in recipients, the leverage has merely shifted. Open standards, local maintenance and multiple suppliers matter.
- **Co-ownership.** Local firms, cooperatives and universities should own the technology, with transparent licensing.
- **Match technology to calories.** Controlled-environment farming serves nutrition and import substitution for high-value foods; staples need irrigation, seed, soil and input strategies.
- **Energy first.** Pair agritech with cheap, ideally renewable, power; unaffordable electricity is the main reason many ventures fail.
- **Beware aid volatility.** Reductions in official development assistance and humanitarian funding in 2025-26 make recipient-owned finance and private co-investment more important.

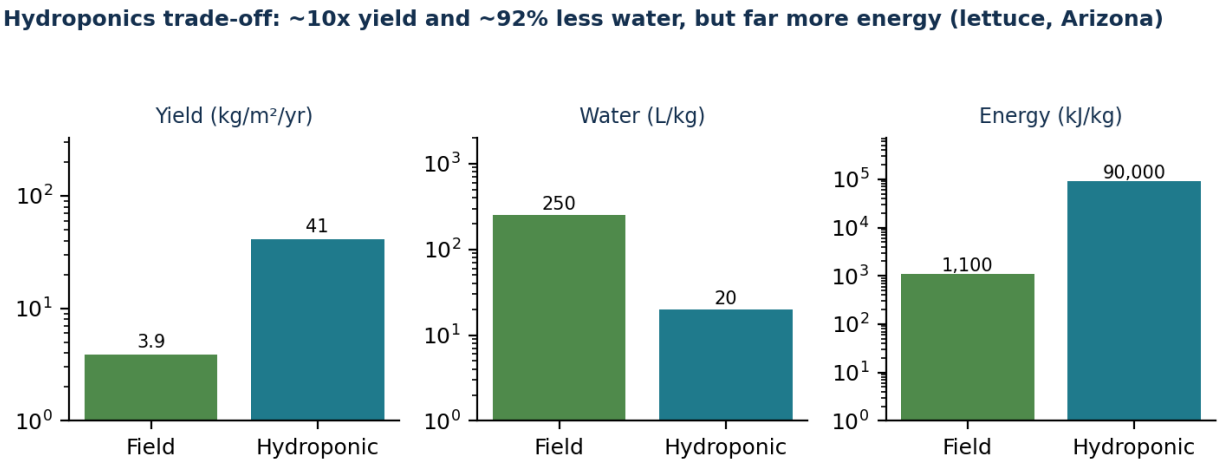
9. Technical Solution Analysis: Hydroponics and Aquaponics

Controlled-environment agriculture (CEA) covers greenhouses, plant factories and vertical farms in which temperature, humidity, light, water and nutrients are managed. The two systems most relevant to resilience are **hydroponics** (plants in nutrient solution) and **aquaponics** (hydroponics coupled with fish culture).

9.1 Hydroponics

Mechanics. Roots are supplied with a water-based nutrient solution, containing nitrogen, phosphorus, potassium, calcium, magnesium, sulfur and micronutrients, delivered to an inert medium (coco coir, perlite, rockwool) or flowing directly over roots. Common designs include nutrient film technique (NFT), deep-water culture (DWC), drip systems and Dutch buckets. Typical management targets are a pH of roughly 5.5-6.5 and an electrical conductivity of about 1.5-2.5 dS/m, adjusted by crop. Sensors and automation maintain the solution; reservoirs recirculate water.

Water efficiency and yield. Because water is recirculated and evaporation and drainage losses are limited, hydroponic crops use a fraction of the water of field production. In a widely cited life-cycle comparison for lettuce (Barbosa et al., 2015), hydroponics produced about ten times the yield per unit area and used over 90% less water per kilogram, but consumed far more energy per kilogram, a gap driven mainly by heating, cooling and lighting. High-technology tomato greenhouses in the Netherlands exceed 60-70 kg per square metre per year; achievable yields in hot-climate operations are typically lower and depend on cooling.



Source: Barbosa et al. (2015), Int. J. Environ. Res. Public Health; log scale; figures rounded. Energy gap narrows with solar power/efficient LEDs.

Figure 6. Lettuce: field vs hydroponic yield, water and energy (Barbosa et al., 2015; log scale, rounded).

Scalability in arid regions. Greenhouse hydroponics scales well where there is access to even brackish or desalinated water, solar power and cooling (evaporative pads, shading, increasingly seawater-cooled greenhouses). The Gulf states, Jordan, Morocco and Egypt have active commercial programmes. Three cautions apply: (i) **economics** are sensitive to energy, labour and market access, and several heavily funded vertical-farm companies went bankrupt between 2023 and 2025; (ii) **crop range** is concentrated in leafy greens, herbs, tomatoes, cucumbers, peppers and berries; staple grains and tubers are not economical; and (iii) **skills and spare parts** must be available locally. Low-tech designs, such as shade-net greenhouses and simple drip-based systems, often deliver the best return in low-income settings. Hydroponic fodder (sprouted barley, ready in about a week) can reduce reliance on imported feed for livestock.

9.2 Aquaponics

The circular ecosystem. Fish (commonly tilapia, catfish or carp) are fed a protein-rich diet and excrete ammonia. Nitrifying bacteria in a biofilter convert ammonia to nitrite and then to nitrate,

which plants absorb as their main nitrogen source. The cleaned water returns to the fish tank. Water is recycled with very low losses, mainly transpiration and small discharges.

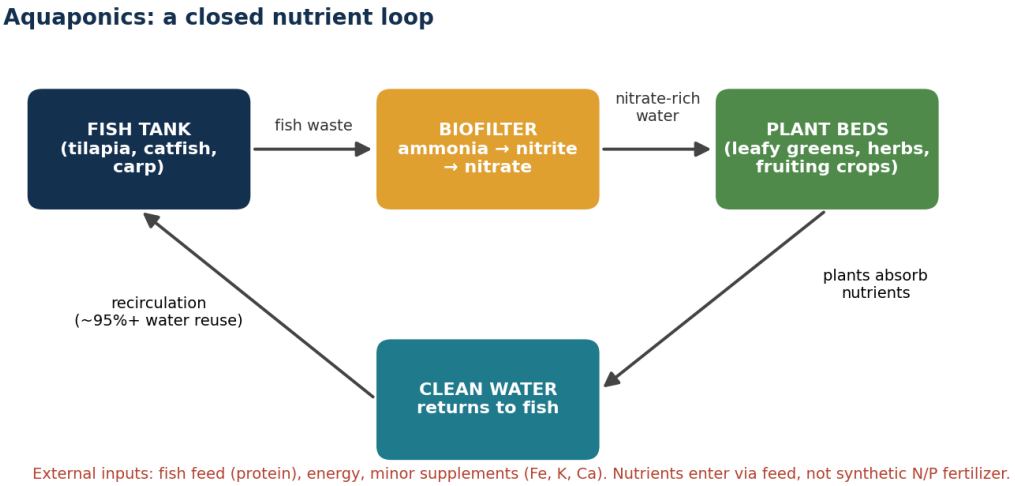


Figure 7. Aquaponics closed nutrient loop.

Nutrient efficiency and benefits. Nitrogen and phosphorus that would otherwise be pollutants in aquaculture effluent become plant nutrients, and the system yields both protein and vegetables from one water volume. It avoids synthetic nitrogen and phosphate fertilizer for the core loop, uses very little water, can be located on marginal or non-arable land, and suits small-scale and community models.

Implementation barriers. The loop is biologically delicate: pH must compromise between fish (about 7), bacteria and plants (about 6), and nutrients such as iron, potassium and calcium often need supplementation. Capital cost and technical skill are significant; fish feed, itself frequently imported and based on fishmeal or soy, is the main external input; power for pumps and aeration is continuous; disease management constrains the use of treatments; and licensing and food-safety rules for integrated systems are underdeveloped in many countries. Crop choice favors leafy greens and herbs, with fruiting crops needing a more mature system.

9.3 Comparative analysis

Dimension	Open-field (baseline)	Hydroponics	Aquaponics
Water use	High; 250 L/kg for field lettuce in one comparison	Very low (about 90% less for lettuce)	Very low; recirculating
Yield per area	Baseline	Several times higher for suited crops	Comparable to hydroponics for greens; lower for fruiting crops
Fertilizer needs	Large synthetic N-P-K	Precise mineral nutrient salts; high use efficiency	Nutrients come via fish feed; minor supplements
Energy	Low	High where heating, cooling and lighting are needed	High, plus aeration and pumping
Capital and skills	Low	Medium to high	High
Main outputs	Staples, fresh produce	Fresh produce and fodder	Vegetables plus fish protein
Exposure to external shocks	Input and weather	Mineral salts, energy, equipment	Fish feed, energy, equipment

9.4 Does widespread adoption reduce dependence on fertilizer monopolies and food imports?

The honest answer is **partly, and indirectly**. Cereals, oilseeds and sugar crops account for the bulk of global fertilizer use, and CEA is not a realistic route to producing these at scale. Hydroponics and aquaponics will not by themselves end dependence on nitrogen, phosphate and potash markets.

What they do offer:

- **Import substitution for perishables.** Fresh vegetables, herbs and fruit are costly to import by air or reefer and are a major micronutrient source. Local production cuts foreign-exchange demand and improves diet quality, which matters because SOFI 2026 reports about 2.7 billion people cannot afford a healthy diet by one estimate.
- **Lower nutrient intensity per kilogram of produce.** Recirculating systems capture nutrients that field systems lose to runoff and leaching, and aquaponics substitutes fish-feed nutrients for synthetic fertilizer.
- **Water and land independence.** Production on marginal land and with brackish or recycled water relieves pressure on scarce aquifers and cropland.
- **Protein security.** Aquaponics adds domestic fish to the diet, reducing reliance on imported animal protein.
- **Shock absorbers and learning platforms.** CEA clusters create engineering, finance and data skills that spill into wider agricultural modernization.
- **Energy linkage.** When powered by solar, CEA reduces exposure to imported fossil fuel; powered by grid gas, it may increase exposure to other chokepoints.

To address staples directly, CEA must be paired with **complementary measures**: soil testing and precision fertilizer application (cutting nitrogen demand 10-20% in many programs), legume rotations and biofertilizers, drought- and heat-tolerant varieties, rainwater harvesting and drip irrigation for open-field crops, strategic buffer stocks, and regional fertilizer production, including emerging green-ammonia projects that decouple nitrogen from imported gas.

Verdict

Hydroponics and aquaponics are **resilience layers**, not replacements for field agriculture. They are most valuable where water is scarcest, import bills are heaviest and cheap clean power is available. Success depends on right-sizing technology, protecting against energy and feed dependencies, and ensuring that local operators can maintain the systems.

10. Conclusion and Strategic Recommendations

Food and fertilizer are now instruments of statecraft because a small number of countries, routes and inputs underpin a large share of global calories. The crises of 2022 and 2026 have different origins, one a deliberate war of conquest and the other the closure of a maritime choke point, yet they impose the same cost on dependent states. Neither multilateral rules nor goodwill will remove this vulnerability soon. The most reliable response is to make dependence less costly to exploit.

10.1 Recommendations for global policymakers (major exporters and donors)

- **Adopt a no-first-use norm for staple food and fertilizer export bans** toward least-developed and net food-importing developing countries, with advance notice and carve-outs for humanitarian purchases.
- **Extend the WFP exemption** to fertilizer shipments for smallholders in crisis countries and to the supporting shipping, insurance and payment channels.
- **Create an Agritech Diplomacy financing window** inside existing development banks, with blended finance for CEA, irrigation, renewable-powered farms and regional input plants.
- **Diversify own supply**: importing OECD and Gulf states should avoid single-route reliance, as Hormuz has shown for fertilizer, sulfur and LNG.
- **Sustain humanitarian and development finance**; cutting it while risks rise raises the long-run cost.

10.2 Recommendations for international bodies

- **WTO:** negotiate binding transparency, proportionality and sunset rules on export restrictions, building on Article 12 and the MC12 precedent.
- **FAO, WFP, IFPRI, AMIS:** maintain open early-warning and export-restriction tracking, now including fertilizer and sulfur flows and shipping through choke points.
- **UN Security Council and ICC/legal community:** operationalize Resolution 2417 with regular famine-risk reporting and clear accountability for attacks on food systems.
- **IMF and World Bank:** make food-and-fertilizer shock facilities standing instruments with fast triggers.
- **Regional organizations (AU, ASEAN, ECOWAS, Arab League, GCC):** build shared reserves, joint purchasing and common standards for agritech trade.

10.3 Recommendations for vulnerable states

- **Diversify suppliers and routes**; qualify at least three origin countries for wheat and two for each major fertilizer, and maintain contingency port and rail options.
- **Hold and manage strategic reserves** sized to the time needed to switch suppliers (typically 90-120 days for cereals), with transparent rules for release.
- **Phase fertilizer-use efficiency**: soil testing, precision application, and organic-mineral blends before any structural change. Avoid abrupt bans (Sri Lanka 2021).
- **Build a CEA and water-smart horticulture base** near cities and in arid zones: shade-net and greenhouse hydroponics first, aquaponics where feed, power and skills allow, always paired with renewable energy.
- **Reform subsidies** gradually towards targeted support to reduce fiscal exposure to price spikes.
- **Negotiate technology partnerships** with local ownership, maintenance training and open standards to avoid new lock-in.

10.4 Phased roadmap

Horizon	Priority actions	Indicators of success
0-12 months	Emergency financing for fertilizer-dependent smallholders; map exposure to Hormuz and Black Sea routes; widen supplier lists; expand export-restriction monitoring	Fertilizer affordability ratio; days of cereal cover; number of qualified suppliers
1-3 years	Launch Agritech Diplomacy window; pilot CEA clusters and training centers; regional blending plants; soil-testing campaigns	Domestic share of fresh produce; yield and water-use metrics; nutrient-use efficiency
3-10 years	Scale renewable-powered CEA and irrigation; regional fertilizer and green-ammonia capacity; binding WTO disciplines; reserve networks	Import-dependence ratios; time-to-recover after shocks; hunger prevalence trend toward SDG 2

10.5 Closing assessment

Hunger is not inevitable: SOFI 2026 shows three straight years of improvement. But the same report warns that conflict, climate extremes and aid cuts could undo that progress, and 2026 has shown how fast an input shock can propagate. The strategic task is to ensure that no single decision in a single capital or at a single strait can hold a nation's food supply hostage. Agritech Diplomacy, backed by reserves, diversification and enforceable norms, is the most practical path to that outcome.

Appendix: Data Notes and Sources

Method and caveats. This report synthesizes publicly available analysis as of 9 October 2026. Where sources conflict (for example on the share of fertilizer trade passing through Hormuz, or on benchmark urea prices), ranges are shown. The FAO price index values, production shares and the vulnerability map in Figures 2, 3 and 4 are rounded or judgement-based and should be verified against original datasets before being cited in formal work. Historical episodes (1973-1980, PL-480, Bangladesh 1974) are summarized from standard scholarship, and causal attributions in some are contested. Events after mid-2026 are changing quickly.

Principal sources consulted

- FAO, IFAD, UNICEF, WFP, WHO (2026). *The State of Food Security and Nutrition in the World 2026*.
- Global Network Against Food Crises (2026). *Global Report on Food Crises 2026*.
- World Bank (June 2026). *Commodity Markets Outlook*.
- IFPRI blog analyses on the 2026 Iran war and fertilizer markets; IFPRI Food and Fertilizer Export Restrictions Tracker (Glauber, Laborde et al.).
- WTO-AXSMarine Strait of Hormuz Trade Tracker (to early September 2026); farmdoc daily (University of Illinois), March 2026; Reuters/Argus reporting on urea and Hormuz, March 2026.
- Mercy Corps (2026). *From Hormuz to the Frontlines of Hunger*.
- USGS Mineral Commodity Summaries (nitrogen, phosphate, potash); FAO Food Price Index.
- IPCC AR6, Working Group II; WRI Aqueduct; FAO AQUASTAT.
- Barbosa, G. L. et al. (2015). Comparison of land, water and energy requirements of lettuce grown using hydroponic vs. conventional agricultural methods. *Int. J. Environ. Res. Public Health* 12(6).
- UN instruments: UDHR (1948); ICESCR (1966) and CESCR General Comment 12 (1999); Additional Protocol I (1977); Rome Statute; UNSC Resolution 2417 (2018); WTO Agreement on Agriculture and MC12 decisions (2022).
- Smil, V. (2001). *Enriching the Earth*.